Module 1 Question bank

Define a random variable. Illustrate the relationship between sample space, random variable and probability.

Define probability theory. Explain conditional probability.

Describe mean, auto correlation and co-variance functions with respect to random process.

Explain the properties of auto correlation function.

Explain the following terms and find the relation between them :
i) Joint probability of events A & B ii) Conditional probability of events A & B.

Define Autocorrelation function. Explain its important properties.

Describe Mean and Covariance function with respect to stationary random process.

What is conditional probability? Prove that $P(B/A) = \frac{P(A/B) \cdot P(B)}{P(A)}$.

With an example, explain what is meant by statistical averages.

What do you mean by probability density function? Prove that the total volume under the surface of a probability density function (pdf) is always 1.

Explain the following terms and find the relation between them :
i) Joint probability of events A & B ii) Conditional probability of events A & B.

What is binary symmetric channel? Obtain a posteriori probabilities for the binary symmetric channel using transition probability diagram.

Define mean, correlation and covariance function of a random process. compute the cross correlation for a pair of quadrature modulated stationary processes $x_1(t) = \cos 2\pi f_c t$ and $x_2(t) = \sin 2\pi f_c t$

Develop a program to generate the probability density function of Gaussian distribution function

Outline random processes and illustrate an ensemble of sample functions with a neat diagram

Show that, if a Gaussian process X(t) is applied to a stable linear filter, then the random process Y(t) developed at the output of the filter is also Gaussian

A random variable is said to be uniformly distributes over the interval (a,b). Determine the probability density function of uniformly distributed RV

Explain central limit theorem as applied to Gaussian Random Process

Determine the characteristic function of a Gaussian random variable with a given mean and variance

Prove the following two properties of the Autocorrelation Function of a Random Process i)If X(t) contains a dc component equal to A, then Rx(t) will contain a constant amplitude equal to A2. ii) If X(t) contains a sinusoidal component, then Rx(t) will also contain a sinusoidal component of the same frequency

Define Autocorrelation Function and Crosscorrelation Function State and prove the properties of ACF

Module 5

Explain the following :
i) Short Noise ii) Thermal Noise iii) White Nose iv) Noise Figure v) Noise Equivalent Bandwidth.

Suppose amplifier 1 has a noise figure of 9dB and power gain of 15dB. It is connected in cascade to the other amplifier 2 with noise figure of 20dB. Calculate the overall noise figure for this cascade connection in decibel units.